

N6Cam+.Compute

Variant: PVT - FULL

11/21/2024

RevB4

CANDIDATE

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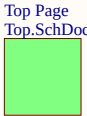
DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational
design notes .

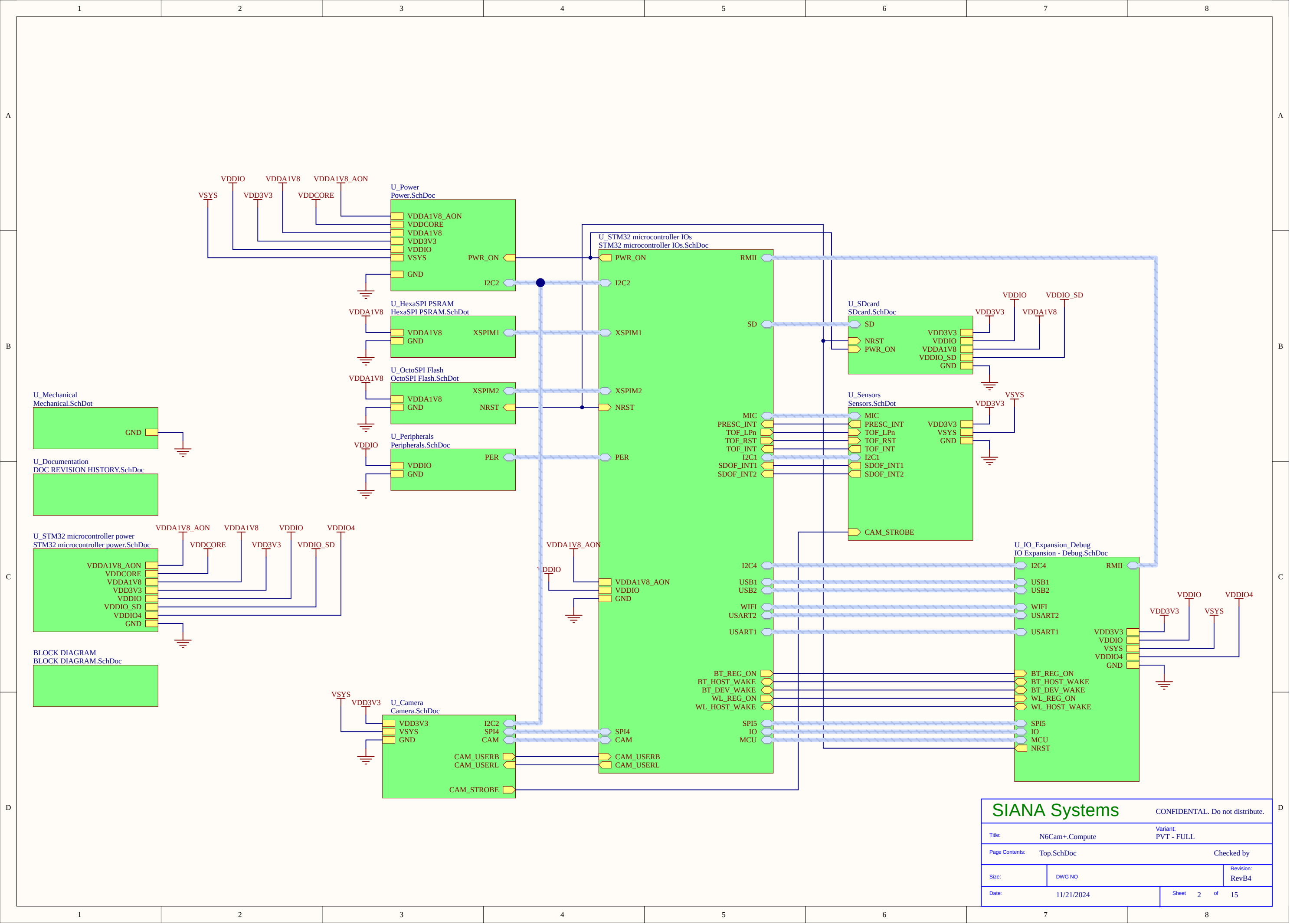
DESIGN NOTE:
Example text for critical
design notes.

DESIGN NOTE:
Example text for cautionary
design notes.

LAYOUT NOTE:
Example text for critical
layout guidelines.



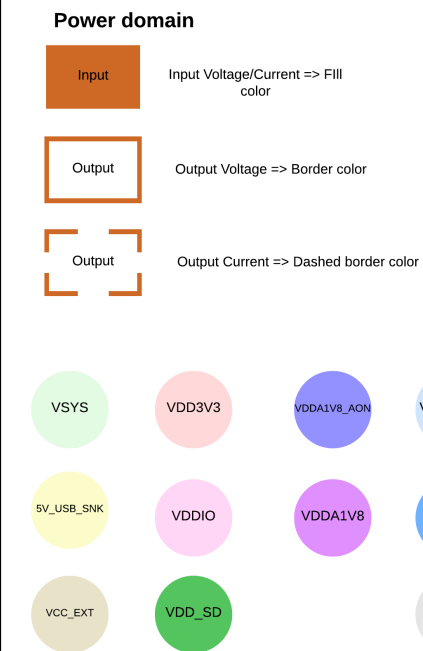
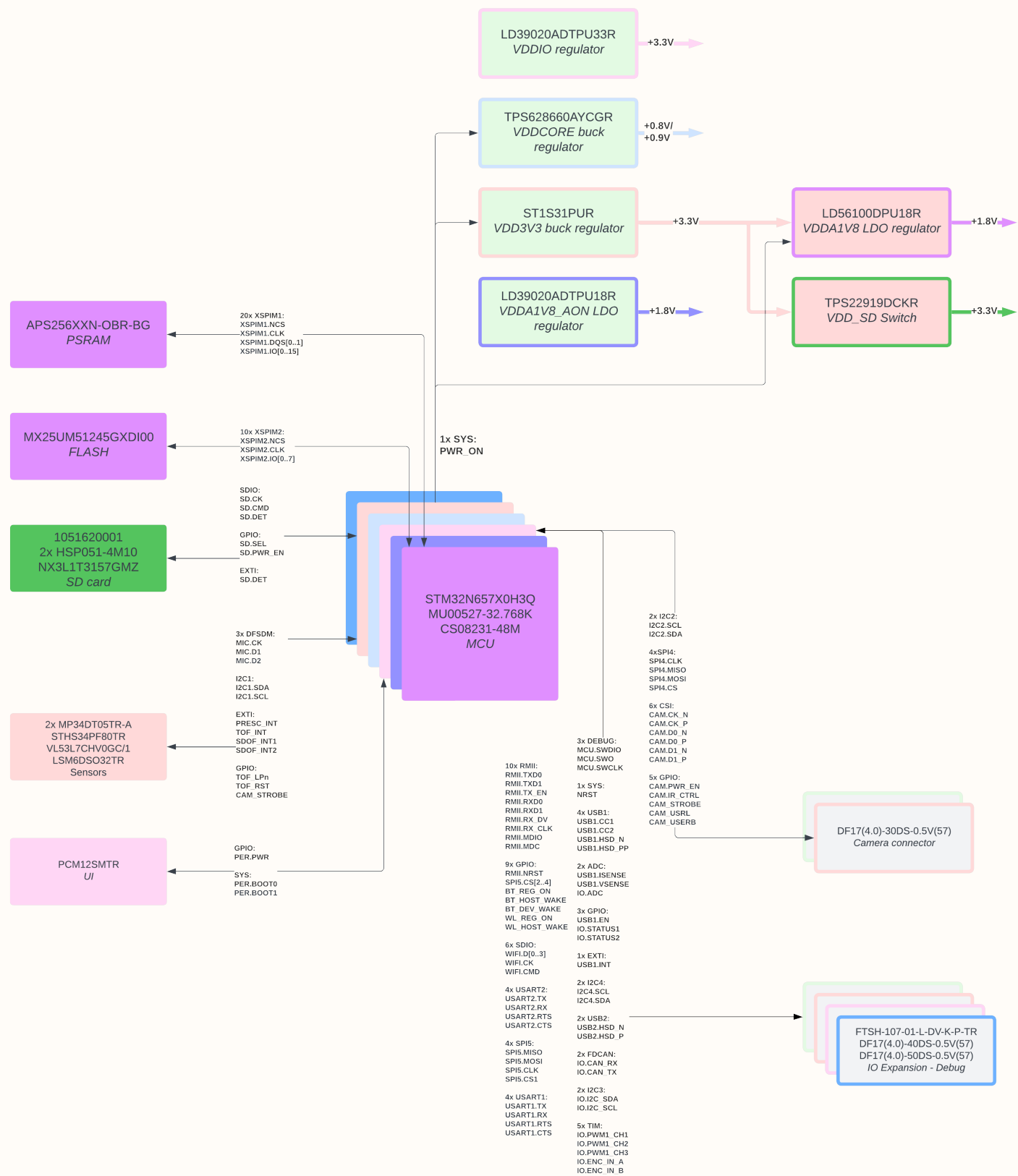
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N6Cam+.Compute

Block Diagram



N6Cam+.Hw

(*) Alternative
(**) Not populated

RevA1
• Initial release

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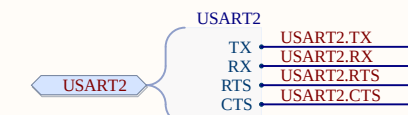
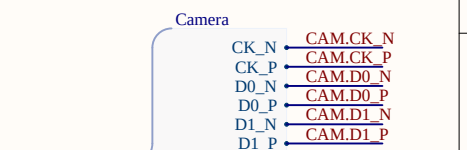
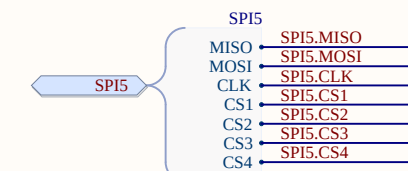
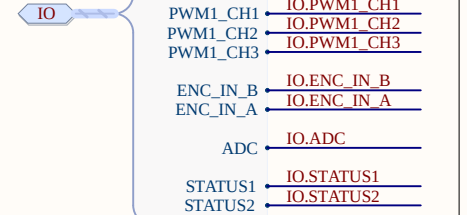
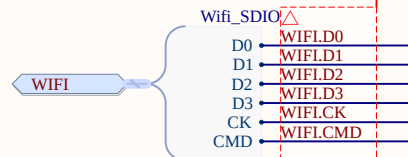
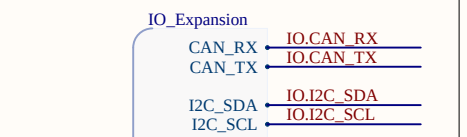
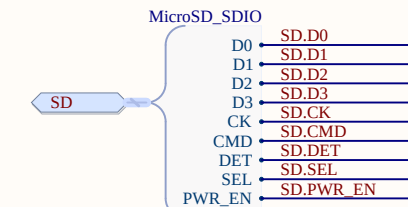
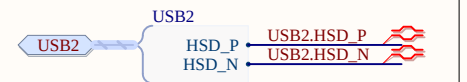
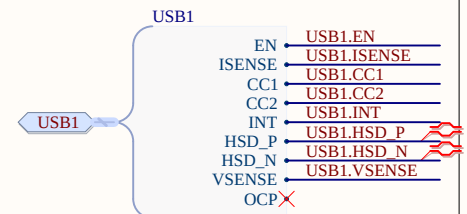
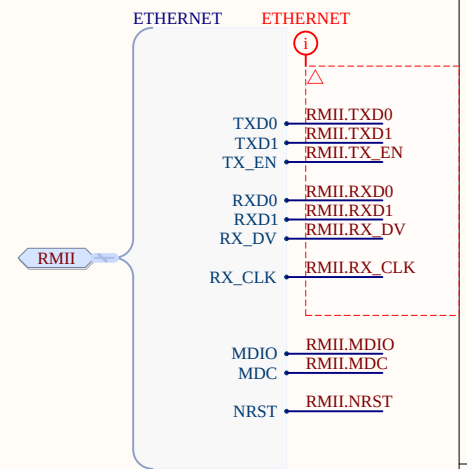
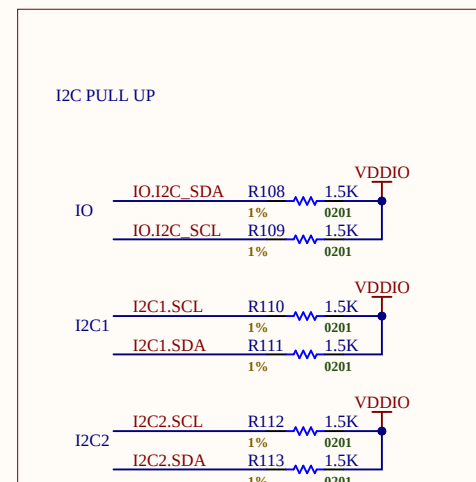
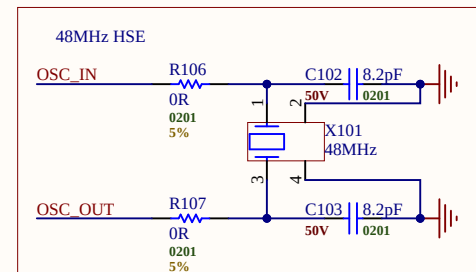
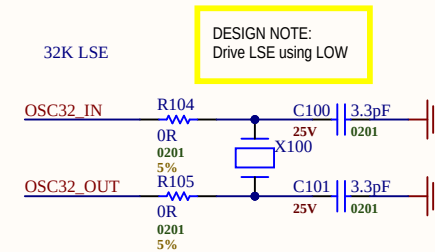
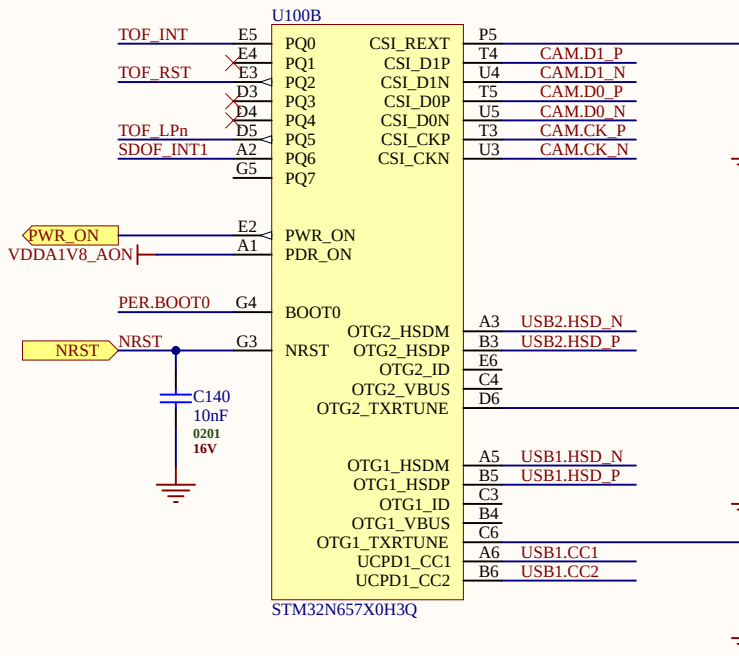
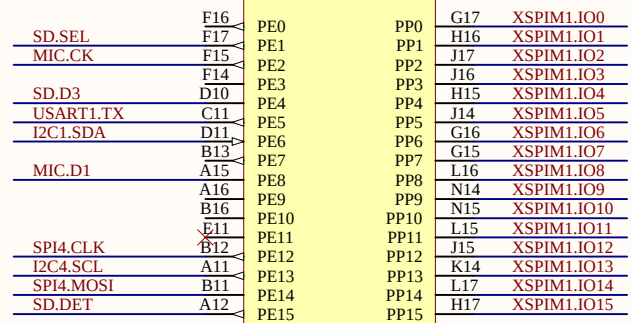
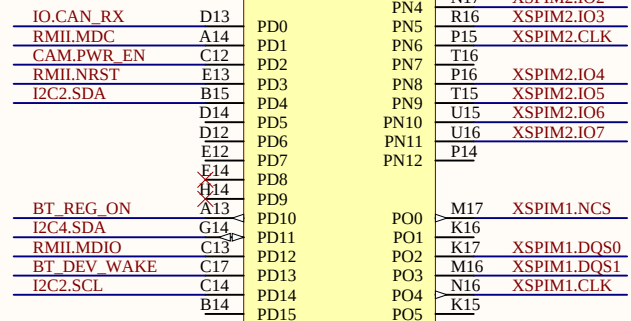
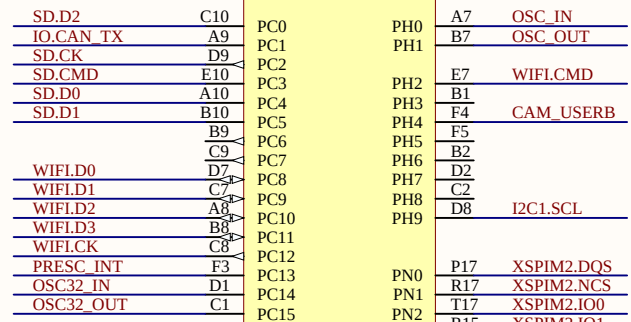
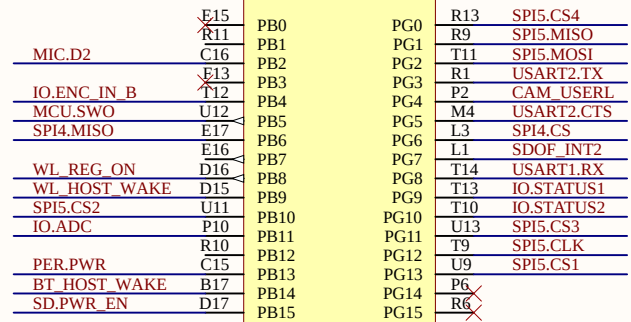
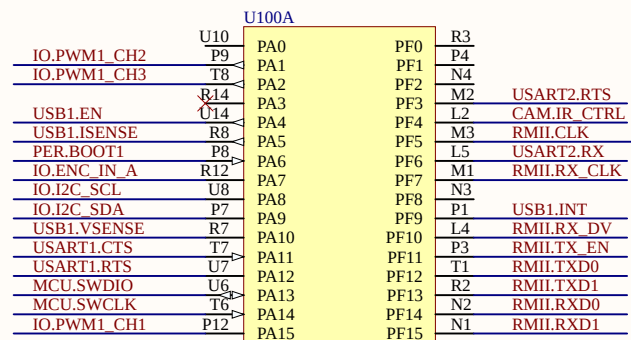
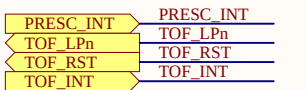
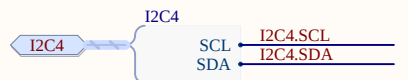
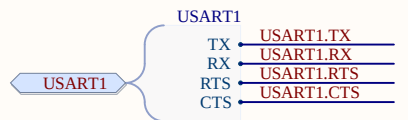
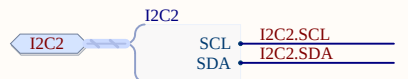
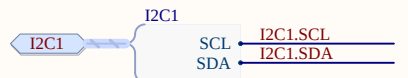
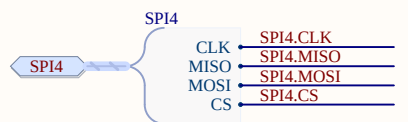
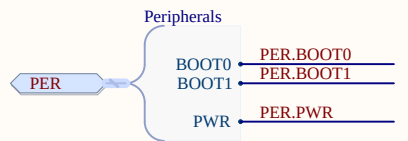
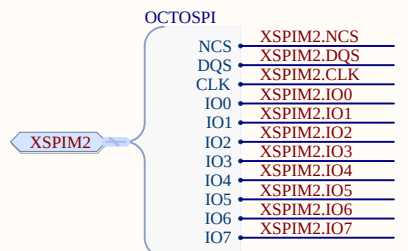
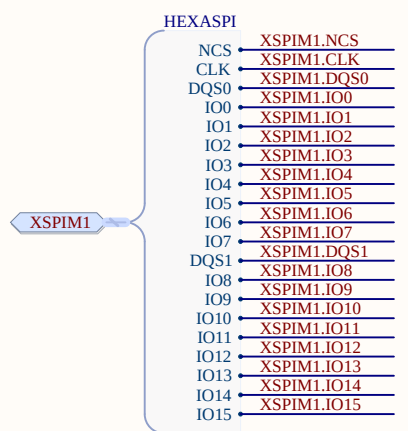
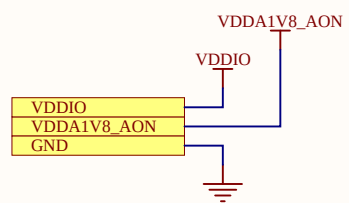
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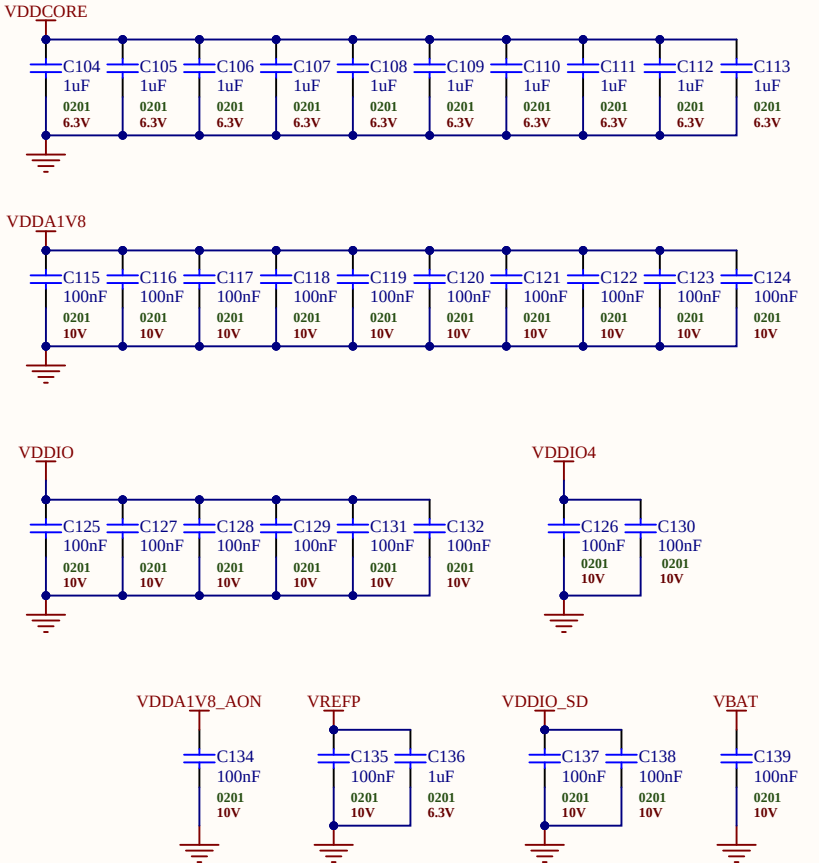
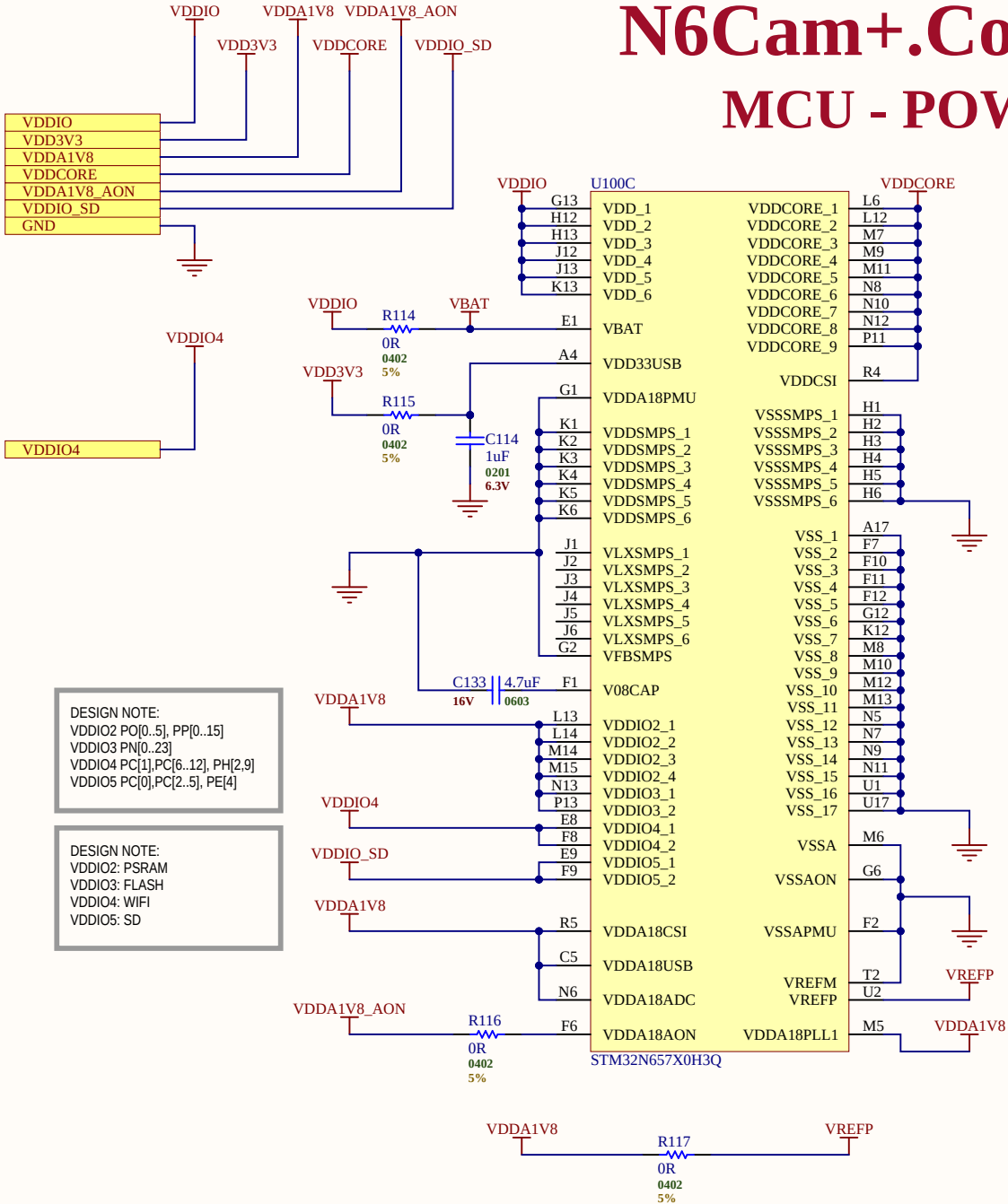
N6Cam+.Compute

MCU - IOs



N6Cam+.Compute

MCU - POWER

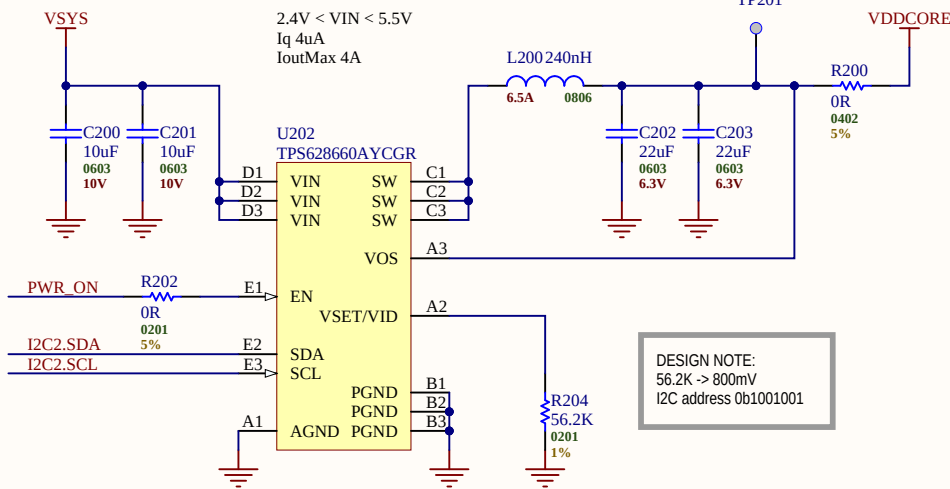


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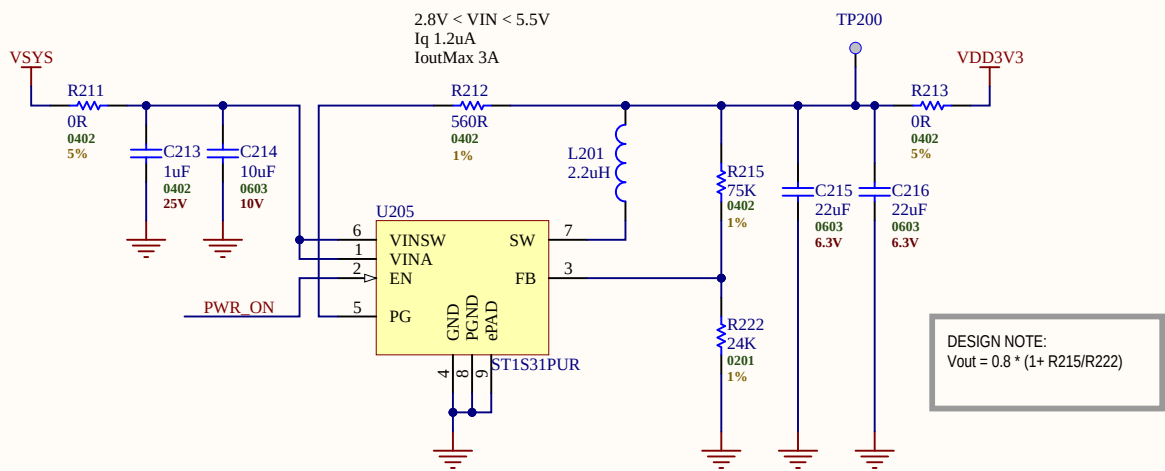
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POWER

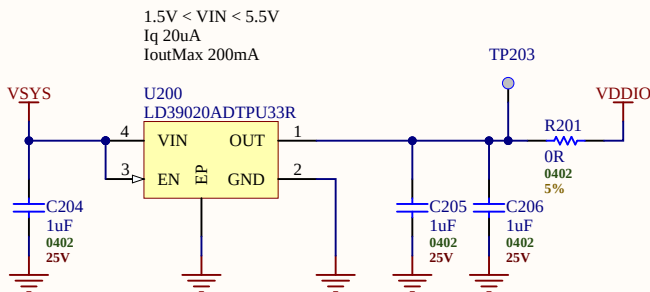
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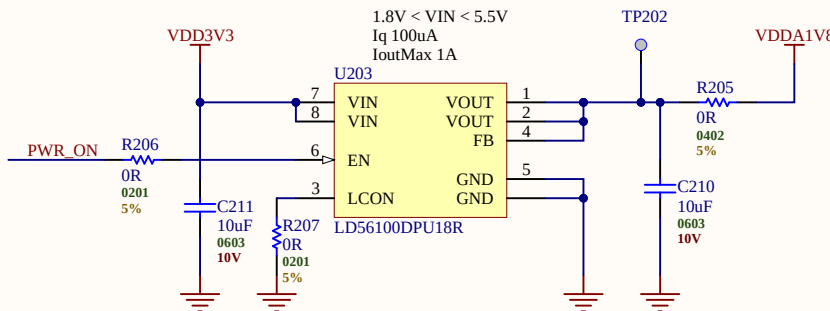
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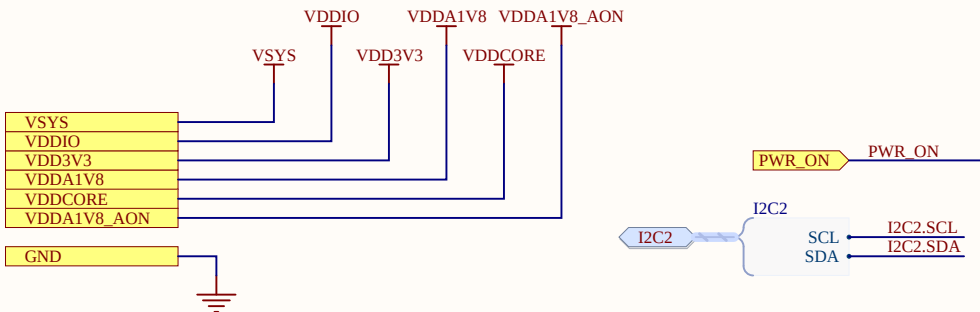
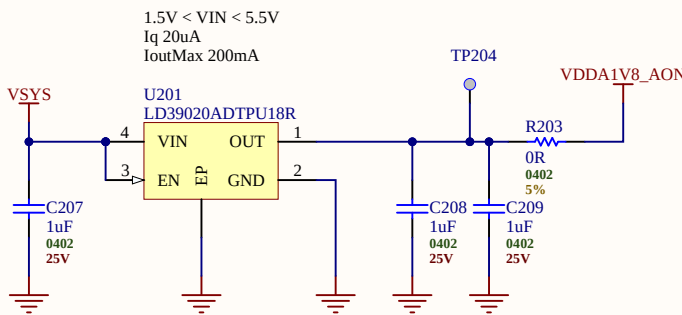
VDDIO



VDDA1V8



VDDA1V8_AON



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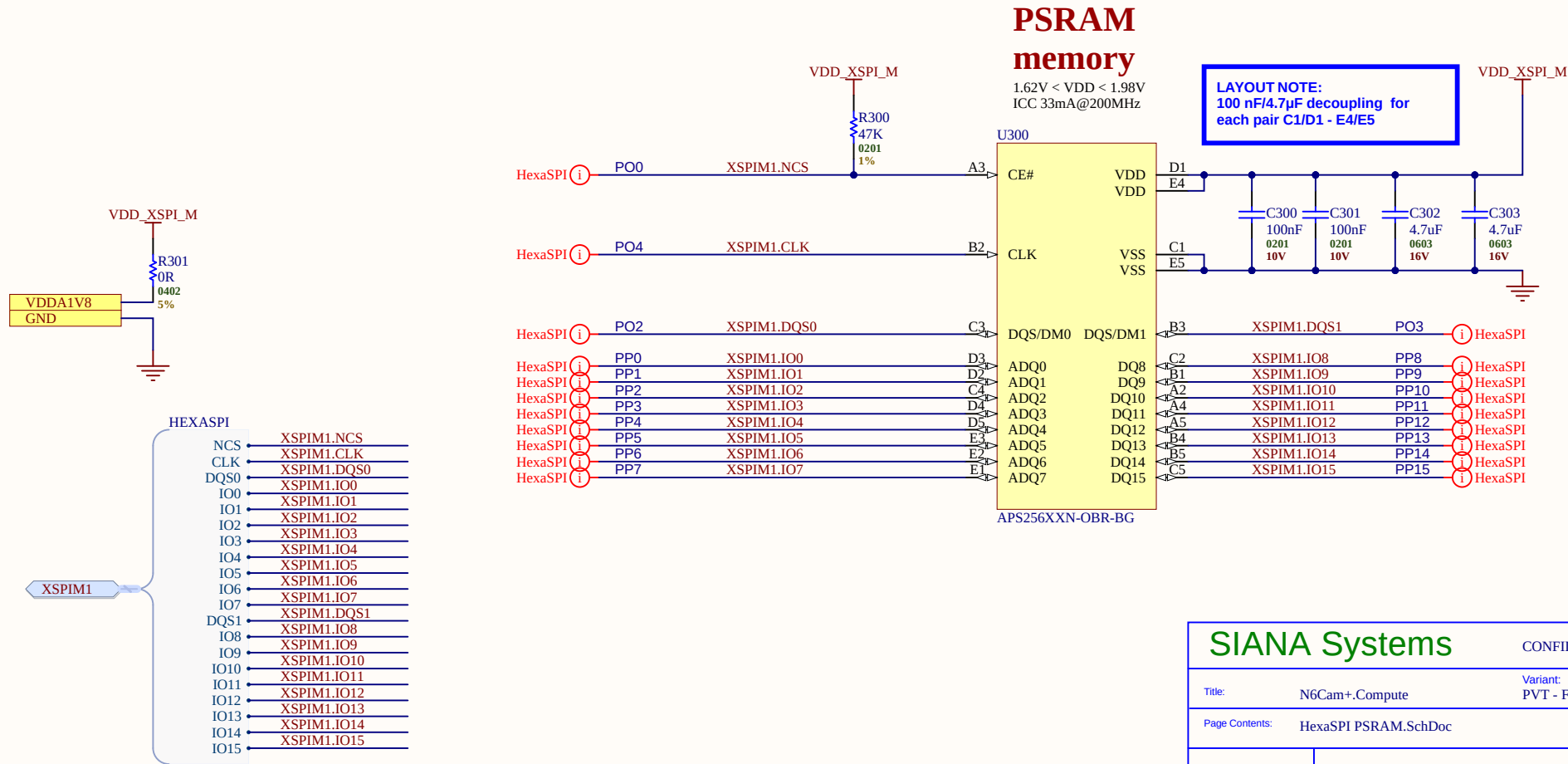
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N6Cam+.Compute PSRAM

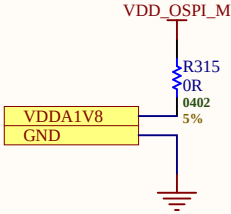
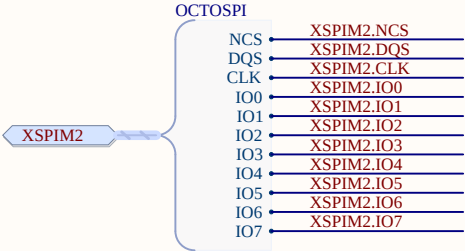
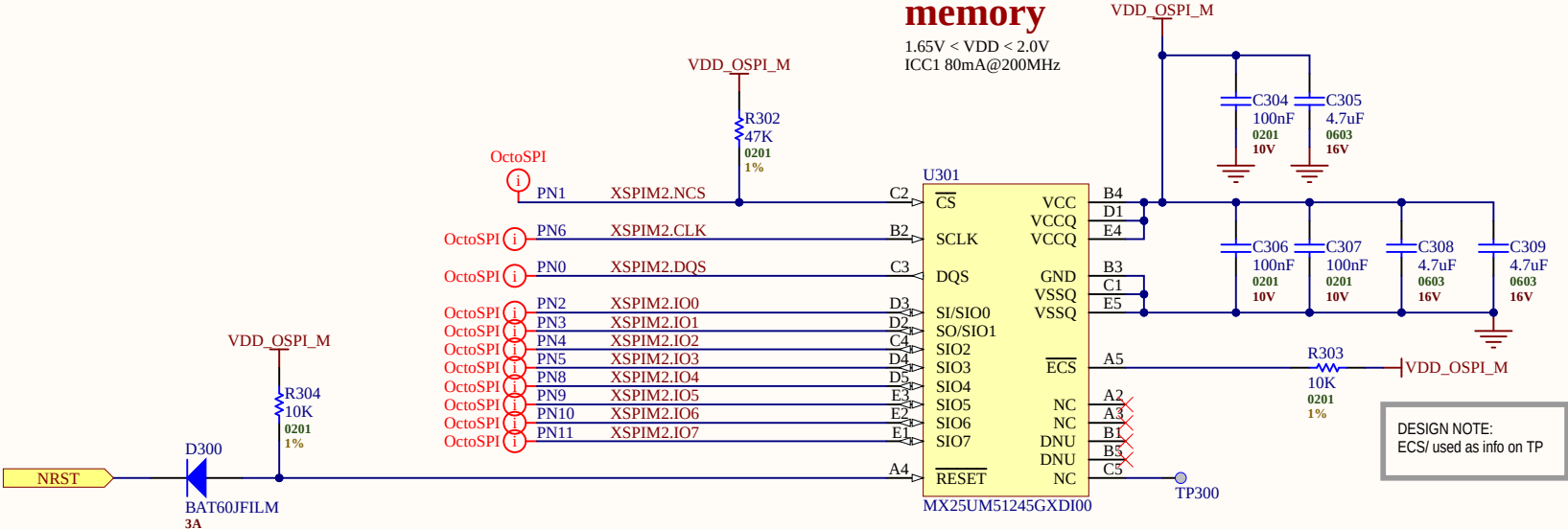


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N6Cam+.Compute FLASH

FLASH memory

1.65V < VDD < 2.0V
ICC1 80mA@200MHz



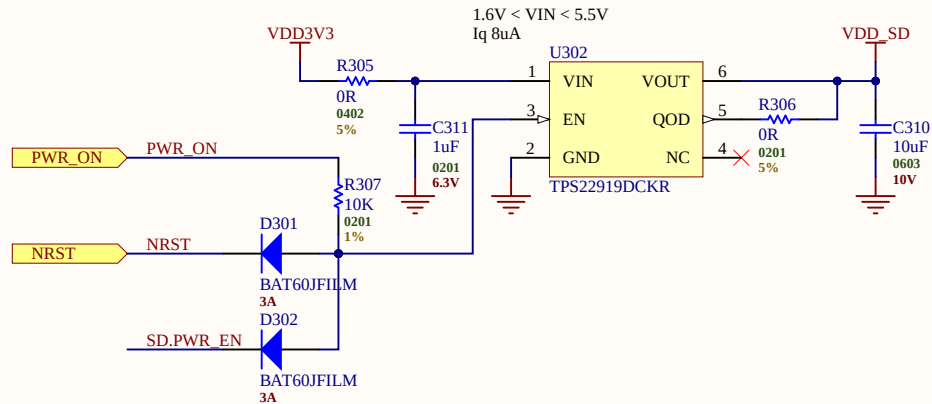
Length equalization (include BGA package length) :

- DQS0 with IO0..7 (Byte0) , and CLK , all in 50 mils
- NCS length can be in the range of up to CLK length +/- 750 mils

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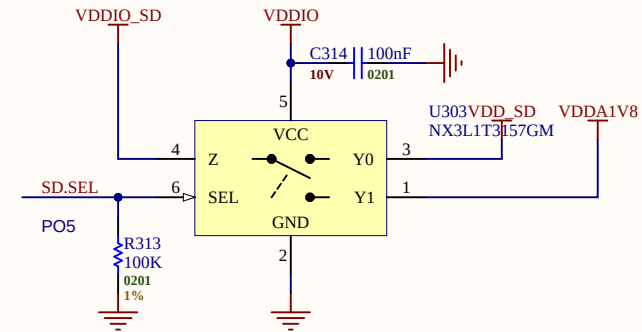
N6Cam+.Compute SD Card

Power switch



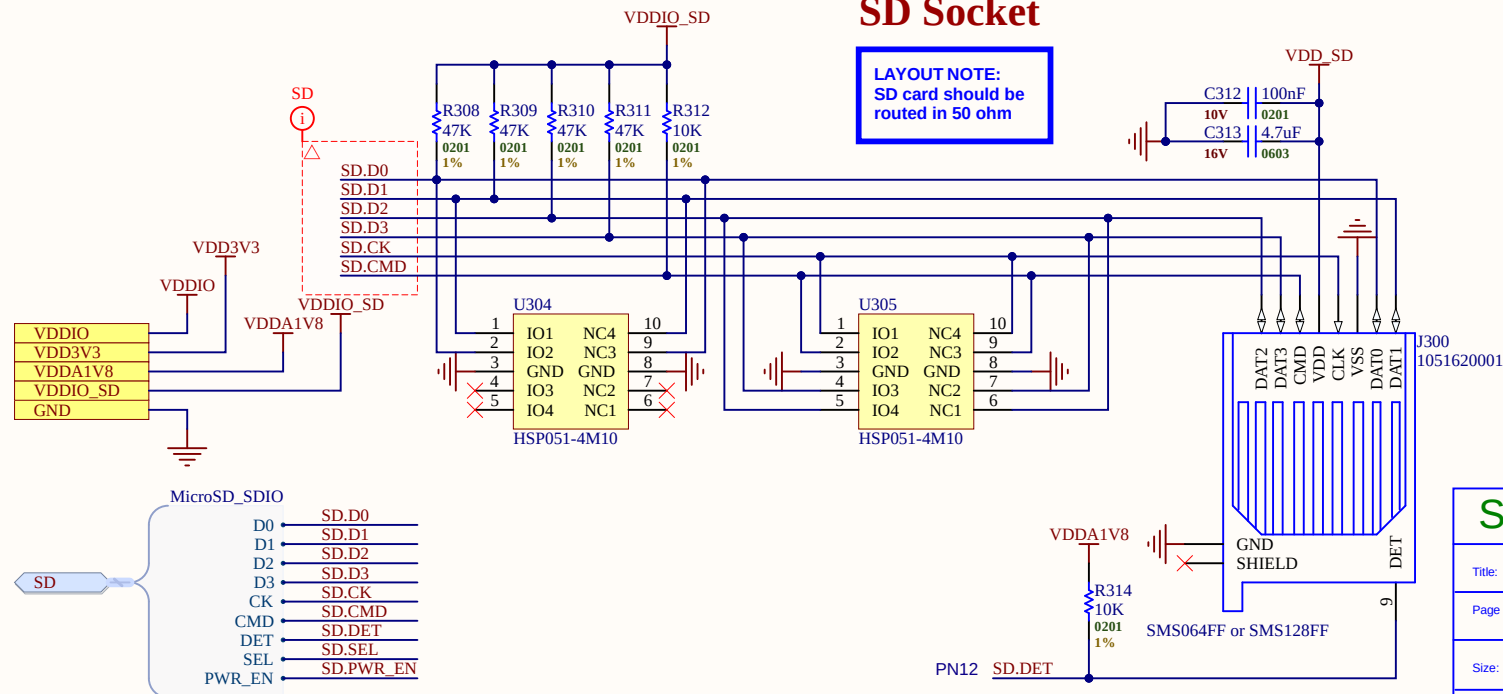
Power multiplexor

1.4V < VCC < 4.3V
Icc 100nA



SD Socket

LAYOUT NOTE:
SD card should be
routed in 50 ohm



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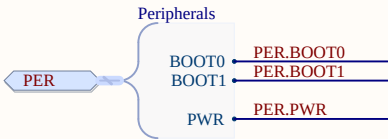
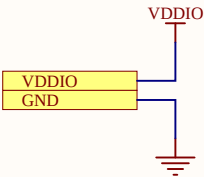
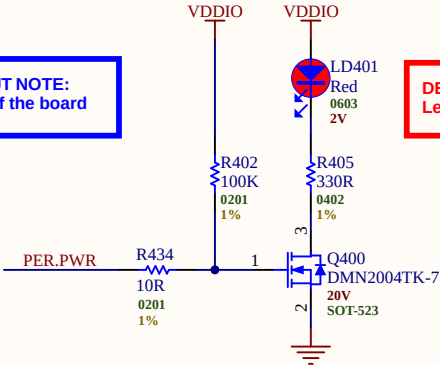
Peripherals

Power LED

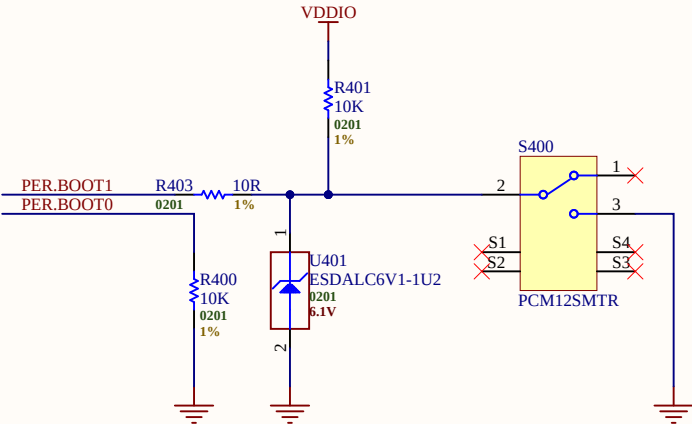
4mA@+3.3V

LAYOUT NOTE:
Back of the board

DESIGN NOTE:
Leave as default ON



BOOT switch



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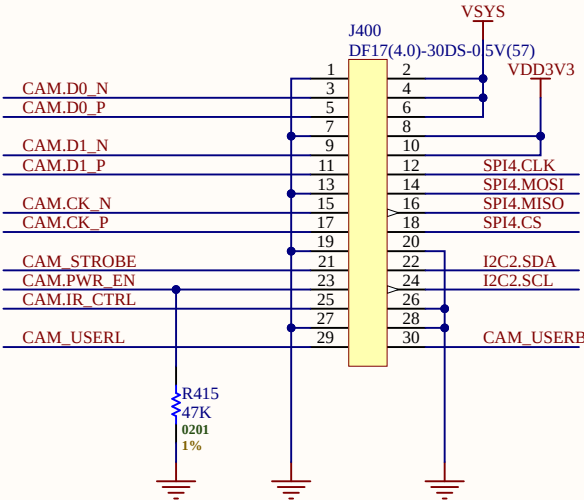
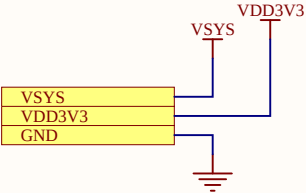
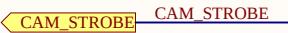
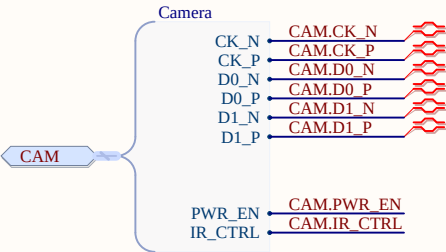
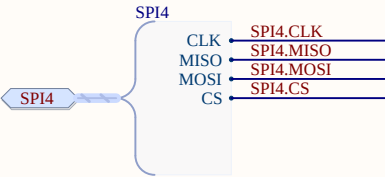
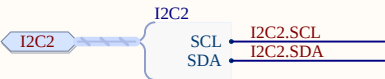
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N6Cam+.Compute CAMERA



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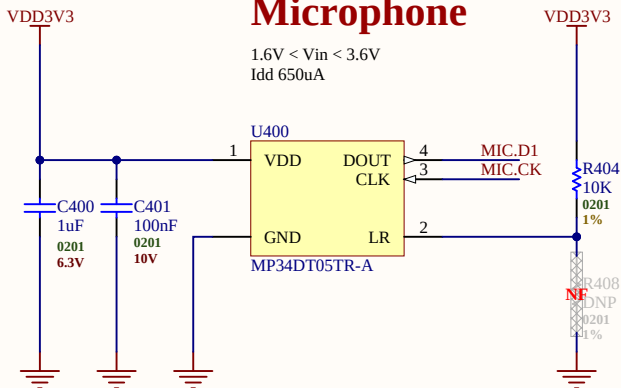
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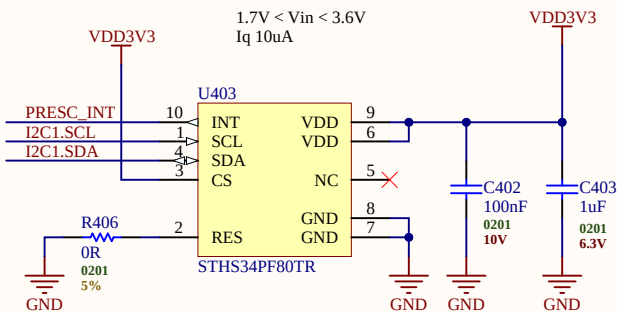
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N6Cam+.Compute Sensors

Microphone

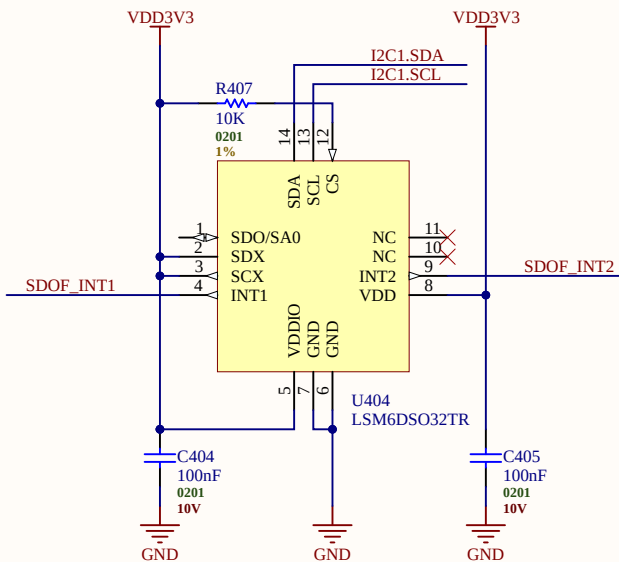


Presence sensor

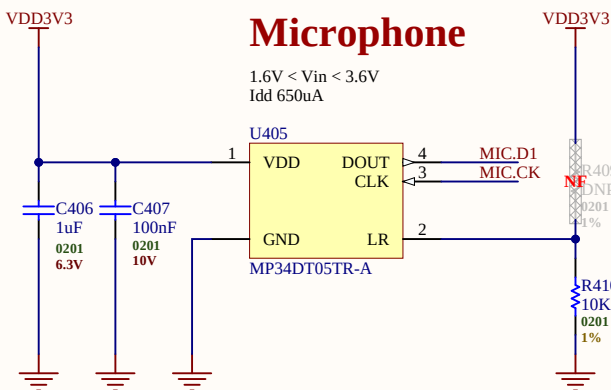


6DOF

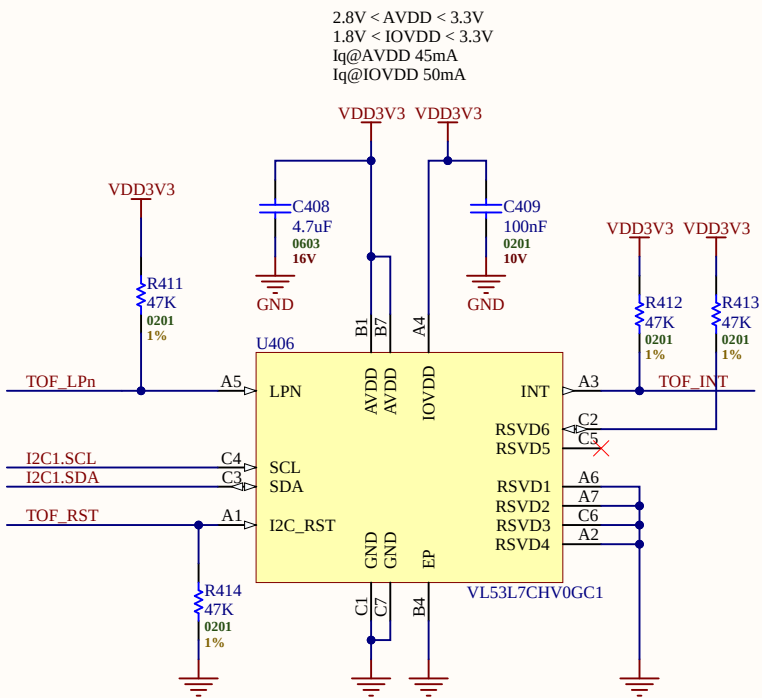
V_{in}: 1.71V to 3.6V
Low-noise Mode:
Gyroscope + Accelerometer current: 0.55mA
Ultra Low-Power Mode:
Accelerometer current: 9.5uA@52Hz, 4.4uA@1.6Hz
Power down mode:
Gyroscope + Accelerometer current: 3uA



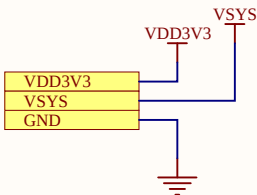
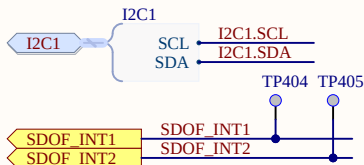
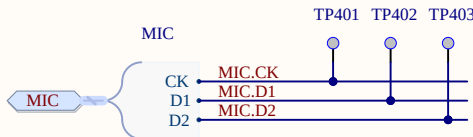
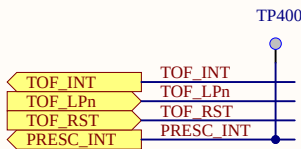
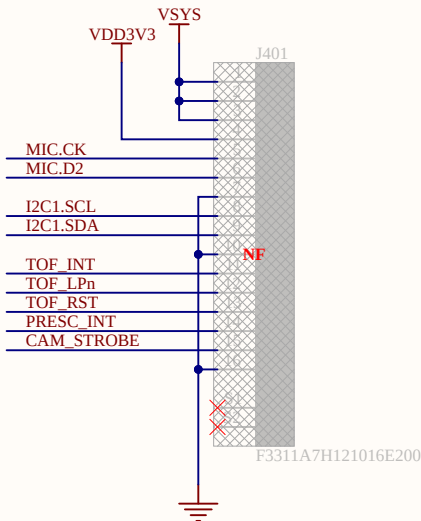
Microphone



ToF



External sensors connector



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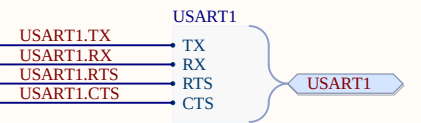
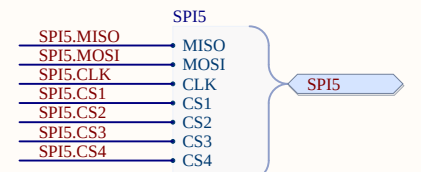
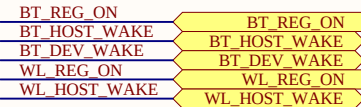
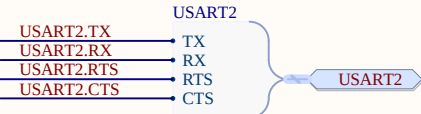
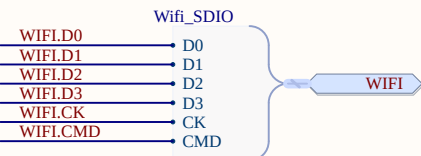
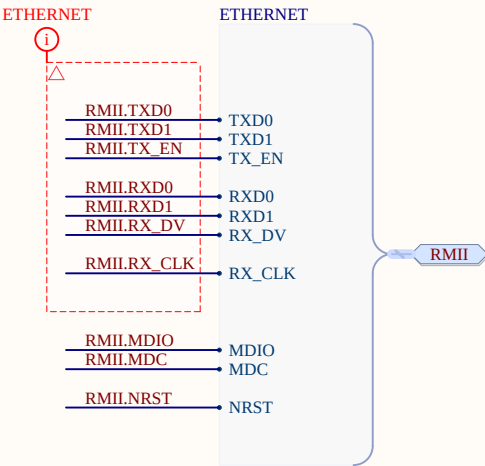
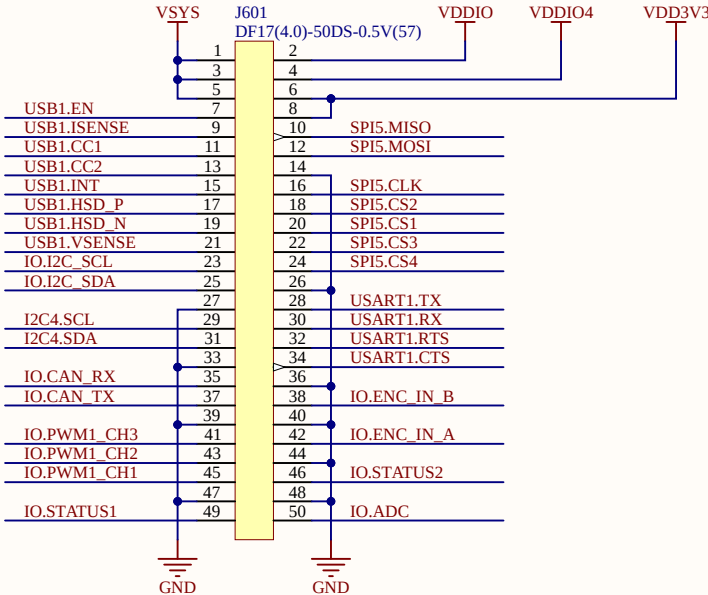
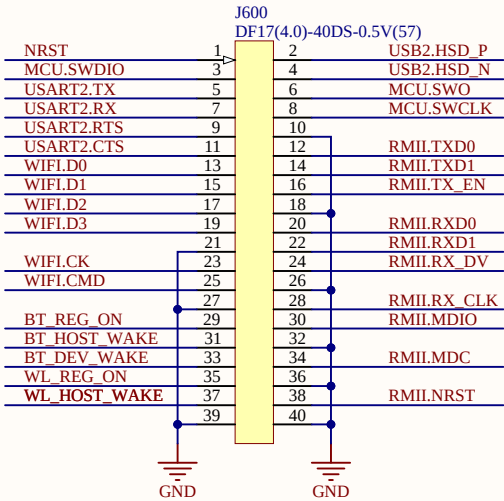
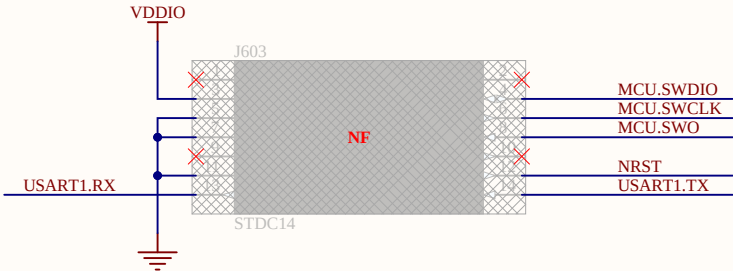
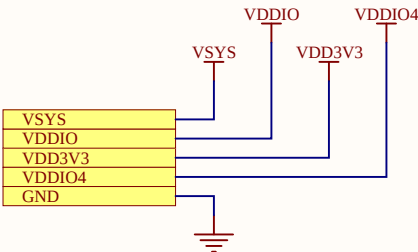
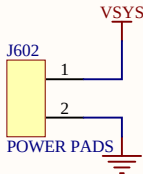
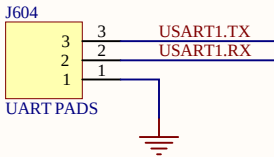
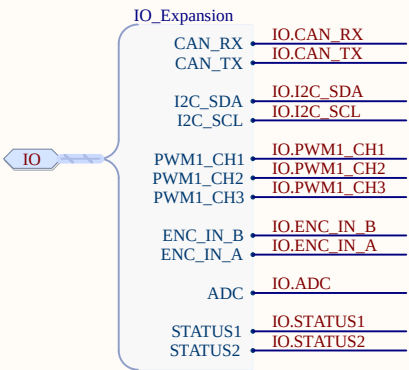
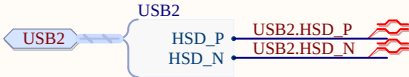
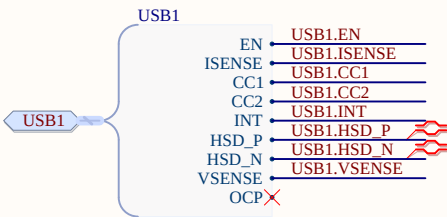
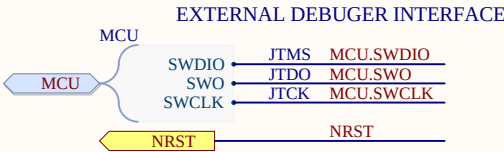
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N6Cam+.Compute

IO Expansion



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N6Cam+.Compute

MECHANICAL

Camera to compute Standoffs



Compute to IO Standoffs



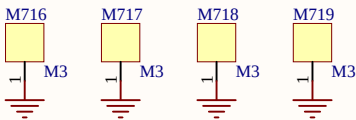
Top screws



Bottom nuts



Mount holes



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DOC: REVISION HISTORY

- RevA1

Initial release
- RevA2

ENGINEERING release for CM quoting
- RevA3

CANDIDATE release
- RevA4

New variant to disable onboard sensors and replace them by FPC connector
- RevB1

Add hardware components
- RevB2

Full ENGINEERING manufacturing release
- RevB3

Increase mounting hole clearance
Move uSD connector 1mm
- RevB4

Update hardware components part numbers
Change layout to fix pad2via and via2via of same net signals

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Title:	N6Cam+.Compute		Variant: PVT - FULL
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